1. A 120 MHz clock CLK with 50% duty cycle is available. Using **Johnson counters** (positive-edge D flip-flops) and AND gates, generate the following outputs:

Output	Frequency	Duty cycle
Y1	30 MHz	50 %
Y2	30 MHz	25 %
Y3	20 MHz	50 %
Y4	20 MHz	33.33 %

   A. Draw the block diagram and the waveforms of CLK, the counter outputs used, and Y1–Y4.
   B. The flip-flops have t_cq = 1.5 ns and t_su = 0.5 ns, and the AND gate delay is 1 ns. For the 2-bit Johnson counter circuit:
      I. Find the maximum CLK frequency at which it still divides by 4 correctly (Y1, 50 %).
      II. Y2 (25 %) drives a flip-flop clocked by the same CLK, so Y2 must be stable before the next rising edge. Find the maximum CLK frequency for the 25 % output.

2. A serial link sends data one bit per clock in **9-bit frames**: 8 data bits followed by an **even-parity bit**, so every correct frame contains an even number of 1s. Design a parity checker that sets **ERROR = 1 for one clock** after any frame with an odd number of 1s.

   A. Using one D flip-flop and an XOR gate, design a circuit whose output Q is 1 when an odd number of 1s has been received.
   B. Design a frame counter that gives FIRST = 1 during the first bit of each frame. State its modulus, the number of flip-flops, and the clear condition.
   C. Add gates so that Q restarts at the first bit of every frame, and write the equation for ERROR.
   D. Trace Q and ERROR for the frames **1011 0010 | 0** and **1101 0110 | 0**. Which frame has an error?
   E. With t_cq = 2 ns, t_su = 1 ns, t_AND = 2 ns, and t_XOR = 3 ns, find the maximum bit rate.